

# Report 5 - Friction Model Registry for OpenHPL/OpenHPLjl

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## Abstract

This report separates hydraulic friction from the RigidPipe momentum equation and introduces a build-time friction-model registry for OpenHPLjl. The objective is to keep the waterway equations reusable while allowing simple quadratic losses, Darcy-Weisbach models, laminar friction, and explicit turbulent correlations to be selected without changing the component architecture.

## 1 Updated Context

The first rigid-pipe model used the compact hydropower loss law

$$H_f = RQ|Q|.$$

This is useful for control-oriented studies, but the coefficient  $R$  hides pipe geometry, roughness, viscosity, and flow regime. OpenHPL includes a Darcy-friction function that distinguishes laminar, transition, and turbulent regimes, so OpenHPLjl now separates friction laws into a reusable function family.

## 2 Generalized Concept Model

The waterway momentum equation remains

$$\frac{L}{gA}\dot{Q} = H_{\text{in}} - H_{\text{out}} - H_f(Q).$$

Only the constitutive relation  $H_f(Q)$  changes. The architecture is therefore

RigidPipe momentum balance + selectable friction law

.

A friction model is selected when the ModelingToolkit system is constructed.

## 3 Other Models Available in the Literature

The first registry contains

none  $\rightarrow$  quadratic  $\rightarrow$  Darcy constant- $f$   $\rightarrow$  laminar  $\rightarrow$  Haaland / Swamee-Jain

.

The implicit Colebrook-White equation is retained as a residual function for later solution with a nonlinear solver. More detailed models can later include transitional blending, unsteady friction, local fitting losses, and frequency-dependent wall friction.

## 4 Mathematics Involved in the OpenHPL Concept/Model

The Darcy-Weisbach head loss is

$$H_f = f_D \frac{L}{D} \frac{Q|Q|}{2gA^2}.$$

The Reynolds number for a circular pipe is

$$Re = \frac{4|Q|}{\pi D \nu}.$$

For laminar flow,

$$f_D = \frac{64}{Re}.$$

The Haaland explicit turbulent approximation is

$$\frac{1}{\sqrt{f_D}} = -1.8 \log_{10} \left[ \left( \frac{\epsilon}{3.7D} \right)^{1.11} + \frac{6.9}{Re} \right].$$

The Swamee-Jain form is

$$f_D = \frac{0.25}{\left[ \log_{10} \left( \frac{\epsilon}{3.7D} + \frac{5.74}{Re^{0.9}} \right) \right]^2}.$$

The Colebrook-White relation is implicit in  $f_D$  and therefore naturally belongs to a nonlinear-solve layer.

## 5 OpenHPL-Specific Modeling Interpretation

OpenHPL's pipe model is based on the momentum equation with friction and gravity forces, and its Darcy-friction helper distinguishes flow regimes. The OpenHPLjl implementation follows the same separation of physical momentum balance from friction-factor calculation while keeping the first implementation explicit and easy to test.

## 6 Implementation in Julia ModelingToolkit / SciML

The friction family is stored independently from waterways:

```
src/FrictionModels/  
|-- FrictionFunctions.jl  
|-- FrictionRegistry.jl  
|-- README.md
```

The registry is

```
const FRICTION_MODEL_REGISTRY = Dict(  
    :none => no_friction,  
    :quadratic => quadratic_head_loss,  
    :darcy_constant => darcy_head_loss,  
    :darcy_laminar => darcy_laminar_head_loss,  
    :darcy_haaland => darcy_haaland_head_loss,  
    :darcy_swamee_jain => darcy_swamee_jain_head_loss,  
)
```

A pipe selects a law at construction time:

```
@named pipe = RigidPipe(
    friction = :darcy_haaland,
    L = 1000.0,
    diameter = 3.0,
    nu = 1e-6,
    epsilon = 1e-4,
)
```

## 7 Model Assembly and Connections

The RigidPipe momentum model is unchanged conceptually:

$$\dot{Q} = \frac{gA}{L} (H_{\text{in}} - H_{\text{out}} - H_f(Q)).$$

Only  $H_f$  is supplied by the selected registry function. This makes the same friction family reusable later in penstocks, tailraces, surge-shaft throttles, and fitting models.

## 8 Numerical Experiment / Validation

For a simple quadratic model,

$$R = 0.5, \quad Q = 2 \text{ m}^3 \text{ s}^{-1},$$

the expected signed loss is

$$H_f = 0.5 \times 2 \times 2 = 2 \text{ m}.$$

For laminar flow at  $Re = 1000$ ,

$$f_D = 64/1000 = 0.064.$$

The unit tests also verify that Haaland and Swamee-Jain produce physically plausible positive Darcy factors for a turbulent test point.

## 9 Expected Outputs

Useful outputs from friction studies are

$$Re(t), \quad f_D(t), \quad H_f(t), \quad \Delta H(t), \quad Q(t).$$

Comparison plots should show head loss versus flow for each registered model and the resulting change in waterway damping.

## 10 Engineering Interpretation

The compact coefficient  $R$  remains valuable for identified or reduced-order hydropower models. Darcy-Weisbach models are more appropriate when geometry, roughness, and viscosity are known. A registry allows these choices to coexist:

$$\text{identified } R \leftrightarrow \text{physics-based Darcy friction}.$$

This is useful for FREKI because friction can later be calibrated from measurements without rewriting the waterway model.

## Conclusion

OpenHPLjl now treats friction as a model family rather than a fixed term. The next step is to compare the registered laws in one RigidPipe simulation and then add transitional and unsteady-friction options where needed.

## References

- [1] OpenHPL, Darcy friction factor function. <https://build.openmodelica.org/Documentation/OpenHPL.Functions.DarcyFriction.fDarcy.html>
- [2] OpenHPL, Pipe model. <https://build.openmodelica.org/Documentation/OpenHPL.Waterway.Pipe.html>
- [3] OpenHydraulics wall friction documentation: Swamee-Jain and Colebrook-White forms. <https://build.openmodelica.org/Documentation/OpenHydraulics.Basic.WallFriction.html>